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Unit – 1 : Session – 9 : SLO – 1 | AND, OR, NOT gates

■ II. Multiple Choice Questions

1. b — All inputs are 1
2. c — All inputs are 0
3. c — $Y = A \oplus B$
4. b — $\oplus$
5. b — 1

■ I. Match the Following

1. AND gate output when all inputs are 1 — 1 (HIGH), from $Y = A \oplus B$ (b)
2. OR gate Boolean expression — a) $Y = A + B$
3. NOT gate symbol — c) $\neg$ (bar)
4. AND gate symbol — d) $\oplus$
5. OR gate output when all inputs are 0 — e) 0

■ III. Activity Question

- AND gate: $Y = A \oplus B$ — output 1 only when ALL inputs are 1. Symbol: D-shape with flat input side. Truth table: 00→0, 01→0, 10→0, 11→1. Example: two series switches lighting one lamp.
- OR gate: $Y = A + B$ — output 1 when ANY input is 1. Symbol: curved input side, pointed output. Truth table: 00→0, 01→1, 10→1, 11→1. Example: doorbell wired to two parallel push-buttons.
- NOT gate: $Y = A \neg$ — output is the inverse of the input. Symbol: triangle with inversion bubble. Truth table: 0→1, 1→0. Example: light that turns ON when a dark sensor reads 0.